

BEFORE CHOICE:

PANOPTICISM AND THE INTERNALIZED REGULATION OF PLAYER JUDGMENT

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ABSTRACT

This study investigates the structural power dynamics of choice-based games by applying Michel Foucault's panopticism to analyze how game systems regulate player judgment. While interactive narratives appear to offer freedom, player behavior often converges toward a perceived correct path. Previous research has primarily explained this through moral identity or reward optimization; however, such post-hoc analyses fail to account for why players fundamentally constrain their own thinking even before a choice is made. This research reinterprets this process as a manifestation of disciplinary power, where system elements such as real-time notifications and flowcharts act as a monitoring gaze. These mechanisms induce a state of constant self-censorship as players align themselves with the system's perspective. However, this study identifies a critical contradiction in existing games: by over-visualizing evaluative criteria, they compromise the invisibility of the gaze, transforming players into strategic analysts rather than disciplined subjects. To address this, the study proposes a design framework that restores the invisibility of power, implemented through the prototype 'Butterfly Effect.' By withholding immediate feedback, the system compels players to further censor and regulate their decision-making attitudes under an invisible evaluative structure. The significance of this study lies in shifting the focus of game choice from branching outcomes to the internalized regulation of behavior under an invisible power structure.

KEYWORDS

Panopticism, Choice-based Games, Interactive Narrative, Feedback Mechanism, Uncertainty in Design

1. INTRODUCTION

Games are a medium through which narratives unfold via player choice. "Choices made during video game gameplay set the stories told in that media apart from other media" (Oliver, 2020). This characteristic is particularly evident in visual novels(hereafter VNs) and interactive movie games, where narrative progression is structured around player choice. Narrative structure in VNs is characterized by overarching 'branching structures'—the macro-level architecture of divergent narrative paths—as well as discrete structural elements such as endings and individual choice points, which serve as the mechanical nodes through which players interact with and influence the narrative (Camingue et al., 2021). Interactive films have been characterized as a form of digital media that allows audiences to influence the fate of characters and engage interactively during viewing, combining cinematic narrative emphasis with game-like participatory structures (Fang et al., 2021). Visual novels and interactive narrative games can be understood as media that place player choice at the core of interaction, granting players the freedom to determine narrative branches and outcomes.

However, in many choice-based games, this freedom gradually converges toward a restricted form of interaction. As "moral decision-making in the context of video games is often tied directly into the game's

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story” (Ferchaud & Oliver, 2019), the engagement and interactivity effects of choice are often attenuated because players perceive a “correct way to play,” frequently assuming that moral decisions are systematically tied to better gameplay rewards (Yin & Xiao, 2022). This phenomenon is empirically supported by Ferchaud and Oliver (2019), whose study revealed that when given a choice, an overwhelming majority of participants opted for the good path, with very few electing to commit immoral actions. This suggests that narrative branching may not always guarantee true autonomy; instead, player behavior tends to be guided by a perceived moral imperative, which can lead them to treat certain choices as incorrect or invalid within the game system.

Choice, therefore, comes to function not as an exploration of experience, but as an object of system learning, through which players infer rules, predict outcomes, and optimize future decisions. Explicit feedback mechanisms encourage players to adjust their behavior according to predefined evaluative criteria. As Whitson (2012) notes, games provide precise real-time feedback that enables players to track progress and determine how to advance, channeling behavior by rewarding actions aligned with system-defined goals while penalizing deviation from them. Kirginas and Gouscos (2018) suggest that the dynamic agency activated by players is inherently dependent on the structural freedom provided by the system, which can often act as a barrier to truly autonomous play. In this way, these systems operate as mechanisms of discipline that regulate the player’s thinking even before a choice is made.

Existing game studies have largely explained this phenomenon through the lenses of morality and meaningfulness (Yin & Xiao, 2022). For instance, Ferchaud and Oliver (2019) found that players’ engagement is deeply tied to their moral identity, noting that an overwhelming majority of players prefer ‘good’ actions to feel more moral and connected to the story. Similarly, studies on ‘morality meters’ suggest that players often rely on visible numerical indicators to make decisions, adopting an instrumental approach to optimize system values rather than engaging in genuine ethical reflection (Ryan et al., 2023). From a more structural perspective, research indicates that game reward systems significantly shape player choices by interacting with other game mechanisms to provide positive player experiences and motivate specific behaviors (Wang & Sun, 2011). Yet such approaches struggle to account for why players so naturally internalize certain evaluative standards and voluntarily regulate their own behavior within these digital structures. This study revisits the issue from the perspective of power, focusing specifically on how thought is structured before a choice is made.

To this end, the study adopts Michel Foucault’s concept of the ‘Panopticism’ as a theoretical framework for game analysis. The panopticon describes a structure in which the continuous possibility of evaluation leads subjects to regulate themselves. As interpreted by Cho (2021), Foucault’s panopticon does not rely on constant observation, but on an asymmetry of visibility through which individuals internalize surveillance and discipline themselves through self-censorship. By applying this concept, this study shifts attention away from the outcomes of choices and toward what players become conscious of prior to choosing.

Throughout this study, the term ‘choice-based games’ refers specifically to visual novels and interactive movie games—genres in which player decisions directly determine narrative branching and outcomes. While broader terms such as ‘games,’ ‘game systems,’ and ‘interactive narratives’ are occasionally used for theoretical generalization, the analysis and design proposals presented here are grounded in these specific genres.

In particular, the study critically examines how explicit feedback structures—common in visual novels and interactive movies—externalize power and position players as agents of optimization. In contrast, it theoretically proposes an alternative design condition, exemplified by the game ‘Butterfly Effect’, in which the possibility of evaluation is maintained while the immediate visualization of outcomes is withheld. The purpose of this study is to explore the structural conditions that position players prior to their choices, proposing how panoptic theory can be translated into game design to illuminate the complex relationship between freedom and control.

2. THEORETICAL BACKGROUND

2.1 Visual Novels and Interactive Movie Genres

Visual novels and interactive movie games have developed as choice-based narrative genres centered on player decision-making. In these genres, “the player’s actions and decisions have an influence on the events of the

story, the dialogue of characters, and other factors of the overall gaming experience” (Yin & Xiao, 2022). Narrative progression in these games is facilitated through diverse mechanics beyond simple dialogue choices. These include narrative progression mechanics such as explicit choice through selection or performance, the discovery of fragmented story segments, and the utilization of in-game systems like resource and time management (Carstensdottir et al., 2019). Within such structures, choices demand responsibility and emotional engagement from players, reinforcing the perception that their decisions meaningfully affect the narrative.

However, as Wolf and Perron (2003) point out, this form of agency is often structured as closed interactivity. Player action is confined within a predetermined framework, where interaction consists of navigating among pre-authored narrative paths rather than generating genuinely new content. The player may know that their freedom is controlled but acts as if it were not, while the interactive narrative seeks to conceal its scripted nature behind an ideology of freedom (Wolf & Perron, 2004).

This closed structure becomes particularly visible in narrative-driven games that present branching outcomes and multiple endings. While players experience their choices as meaningful, the underlying logic of progression is fixed in advance by the designer. As Carstensdottir et al. (2019) suggest, this process establishes a form of “common ground” between designer and player, wherein players gradually learn to decode the operational logic embedded in the system in order to progress the story. Player agency thus operates not as open-ended authorship, but as successful alignment with the narrative paths already inscribed within the system.

2.2 Games and Experiential Understanding

While game experiences are guided by the game’s narrative and structure, they are ultimately constructed through the behavioral choices of each user (Kim & Lee, 2016). In games, players interact with systems where rules function not merely as formal qualities, but as a “context designed to deliver a particular experience” (Salen & Zimmerman, 2004). Through this interaction, they experience not only the final outcomes but also the intricate, contextual processes of judgment and decision-making that lead to them. Therefore, rather than a mere collection of rules, games function as experiential media where the ‘alterbiography’ is generated through the player’s “here and now” interactions with the system’s coded structures and entities (Calleja, 2009).

From this viewpoint, games can function as media that allow players to internalize abstract concepts or norms through action. This process of internalization is facilitated by specific structural features of the game system. As Czauderna and Budke (2020) illustrate, techniques such as the curation of possible decisions and the offering of lucid feedback effectively structure the player’s practice of decision-making. This feedback mechanism is central to what Whitson (2012) describes as the “quantification of the self,” where the collection and analysis of minute data provide “feedback on how to better care for one’s self.”

With the game focusing too much on feedback, it gives the player too much to think about and increases cognitive load (Rogers, 2017). This was likely because feedback is meant to funnel players’ behaviors and actions (Rogers, 2017). As Whitson (2013) observes, games excel at rendering abstract progress into visible data, using status gauges and progress bars to create a precise real-time feedback loop that effectively governs player behavior. Consequently, players can gradually learn the causal relationships between choices and outcomes through repeated play. Through this process, players implicitly begin to calculate which choices are most rational or beneficial, continuously gathering and evaluating in-game information in order to select actions that promise optimal outcomes (Kim & Lee, 2016). In this process, the interiority of the self is transformed into an actionable database, leading players to treat their decisions as objects of calculation for optimizing performance and outcomes (Whitson, 2013).

In this sense, the learning enabled by game systems can be reinterpreted as a form of disciplinary power. As players subject themselves to this quantification to “know and master the self,” they find a certain “pleasure associated with this mastery and control” (Whitson, 2013), leading them to voluntarily align their behavior with the system’s gaze even before a choice is made.

2.3 Foucault and Panopticism

The panopticon was not originally conceived by Foucault, but by Jeremy Bentham, an English philosopher and social theorist in 1785 (Liagkou et al., 2021). It refers to a circular prison intended to enable efficient management and surveillance with minimal cost (Cho, 2021). Foucault (2008) describes the architectural figure of this composition as follows:

“We know the principle on which it was based; at the periphery, an annular building; at the centre, a tower; this tower is pierced with wide windows that open onto the inner side of the ring; the peripheric building is divided into cells, each of which extends the whole width of the building; they have two windows, one on the inside, corresponding to the windows of the tower; the other, on the outside, allows the light to cross the cell from one end to the other.”

By this arrangement, the observer can see the inmates while the inmates cannot confirm the presence of the observer. As Foucault (2019) further notes, “The Panopticon is a machine for dissociating the see/being seen dyad: in the peripheric ring, one is totally seen, without ever seeing; in the central tower, one sees everything without ever being seen.” The asymmetry of visibility is not limited to the binary relationship of seeing and being seen; rather, it exerts a profound effect where inmates, under the constant assumption of surveillance, begin to self-censor and internalize discipline, eventually becoming habituated to the operations of disciplinary power (Cho, 2021).

“In the '70s, Foucault studied, deeply, Bentham’s concepts, pointing to the Panopticon as a generic model that denotes a way of defining and discussing power relations in terms of their impact on people’s everyday life” (Liagkou et al., 2021). For Foucault, the essence of the Panopticon lies not in the actual presence of a watcher, but in the internalization of the awareness that one is constantly visible. As Foucault (2008) argues, this asymmetrical gaze induces a state of conscious and permanent visibility in the inmate, ensuring the automatic functioning of power and rendering its actual physical exercise unnecessary through its internal perfection.

Consequently, panoptic power operates through self-surveillance rather than direct physical coercion. By inscribing the power relation within himself, the individual “becomes the principle of his own subjection” (Foucault, 2019). Thus, individuals adjust their behavior and speech by assuming a potential judgment, even without clearly identifying the evaluative criteria. In this process, the Panopticon becomes an experiential structure where individuals transform into subjects who monitor themselves, rendering the exercise of power continuous, automatic, and independent of any specific person (Foucault, 2008).

3. INTERACTIVE NARRATIVE GAMES AND THE PANOPTICON

3.1 Choice Structures in Existing Interactive Narrative Games

Representative examples of interactive narrative games include ‘Until Dawn’ (Supermassive Games, 2015), ‘The Walking Dead’ (Telltale Games, 2012-2019), and ‘Detroit: Become Human’ (Quantic Dream, 2018). These games adopt structures in which accumulated player choices influence narrative progression and endings, with choice functioning as a central mechanism for narrative intervention. ‘Until Dawn’, developed by Supermassive Games, employs horror conventions such as abjection, moral judgement, and suspense, while systematically visualizing key player choices and the consequences of decisions and mistakes to intensify participatory engagement (Fawcett, 2020). Like many interactive dramas, Telltale’s ‘The Walking Dead’ employs a game event tree that directs players along a branching narrative, resulting in a highly scripted structure (Moallem & Raffe, 2020). ‘Detroit: Become Human’ is often categorized as an action-adventure or interactive movie–style game, characterized by a branching narrative structure in which endings are determined by the player’s choices, supported by relatively simple control mechanics and choice-based interactions (Park, 2022). These games adopt structures in which accumulated player choices influence narrative progression and endings.

Such games typically share several features. First, the outcomes of player choices are visualized at the level of events. Specific choices lead to clear narrative consequences such as character survival, relationship changes, or scene transitions, allowing players to easily recognize the effects of their decisions.

Second, these games utilize explicit interface elements to provide real-time evaluative feedback, as shown in the various status and notification systems of major titles. For instance, ‘Until Dawn’ monitors the shifting interpersonal affinity between characters—driven by player-led betrayals or rescues—through the dedicated status screen depicted in Figure 1(a). ‘The Walking Dead’ employs on-screen notifications to signal when an NPC remembers a player’s choice, a mechanism illustrated in Figure 1(b). ‘Detroit: Become Human’ further

reinforces this transparency by displaying relationship updates in the upper-right corner as shown in Figure 1(c), which triggers whenever a player adopts a supportive or oppositional stance.

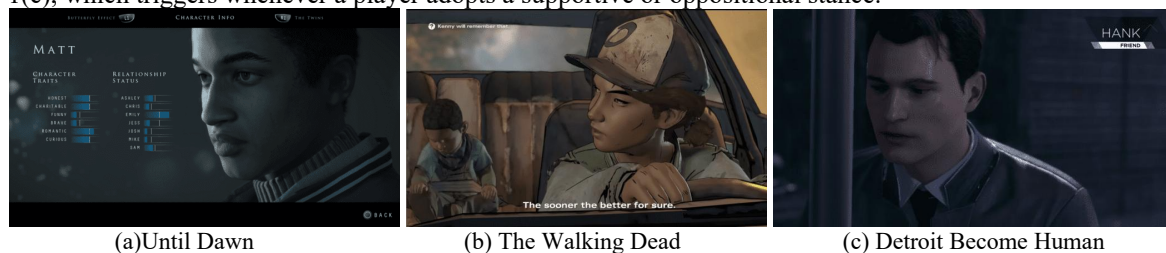


Figure 1. Character relationship status screen in ‘Until Dawn’ (Supermassive Games, 2015) (a), Choice-based narrative foreshadowing in ‘The Walking Dead’ (Telltale Games, 2016) (b), Relationship change notification in ‘Detroit: Become Human’ (Quantic Dream, 2018) (c)

Third, the inclusion of complex branching trees and flowcharts allows players to decode the causal relationships between choices and outcomes through repeated play. A representative example is found in ‘Until Dawn’, which explicitly tracks the consequences of player actions through the ‘Butterfly Effect’ system. This mechanism, as depicted in Figure 2(a), provides an organized summary of interconnected events triggered by the player’s past decisions on a dedicated screen. ‘Detroit: Become Human’ presents a comprehensive flowchart at the conclusion of each chapter to reveal the exact narrative path taken. By displaying locked alternative paths alongside the player’s chosen route, the flowchart shown in Figure 2(b) clearly illustrates where and how narrative branches diverged. While these systems validate player agency, they simultaneously render the underlying system logic entirely transparent.

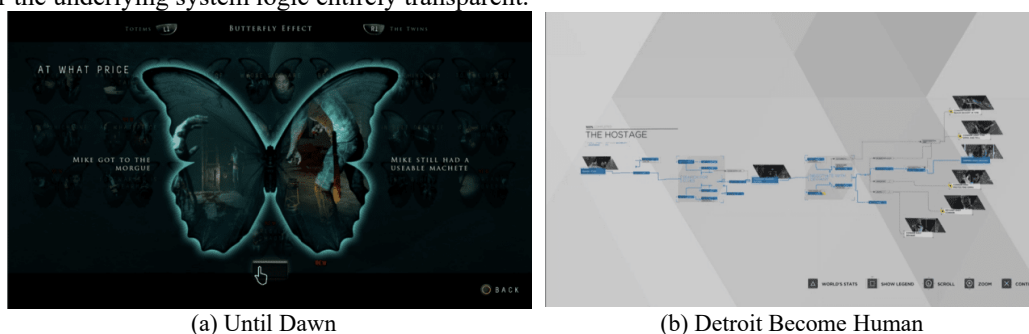


Figure 2. Visualization of event-based causal relationships in ‘Until Dawn’ (Supermassive Games, 2015) (a), Flowchart displayed after chapter completion in ‘Detroit: Become Human’ (Quantic Dream, 2018) (b)

3.2 Interactive Narrative Games and Panoptic Perspective

From a panoptic perspective, existing interactive narrative games establish a structural environment that facilitates the surveillance and regulation of the player. These games align with the foundational conditions of panopticism by fostering an awareness that player actions are constantly recorded and evaluated. Titles such as ‘Until Dawn’, ‘The Walking Dead’, and ‘Detroit: Become Human’ repeatedly remind players that their choices are stored as data, mirroring the power of documentation. As Foucault (2008) illustrates, this disciplinary mechanism functions as a “compact model in which all events are recorded, in which an uninterrupted work of writing links the centre and periphery.” In the context of interactive games, this “uninterrupted work of writing” is manifested through the continuous data-tracking of player interactions, where even the slightest movements are supervised and fixed into a record.

This constant visibility prompts a psychological shift where the external gaze of the game system is gradually internalized by the player. Following Foucault’s (2008) description of a system “which use instruments that render visible, record, differentiate and compare,” interactive games employ interfaces like status screens and logs of outcome changes resulting from choices to compel players to constantly reflect on their actions. As Foucault (2008) defines the essence of panoptic effect, the player as a subject “inscribes in himself the power relation in which he simultaneously plays both roles; he becomes the principle of his own subjection.” Rather than simply playing, they begin to morally self-censor and regulate their behavior in anticipation of the potential consequences of their choices. This signifies that the player, under the persistent

pressure of an evaluative eye that they have accepted as a constant presence, comes to voluntarily regulate their own thought processes even before a choice is made.

However, a critical discrepancy arises between these games and true panopticism due to the excessive transparency of their evaluative criteria. Due to the asymmetry of visibility (Cho, 2021) inherent in the structure of the Panopticon, the subject cannot know the exact moment or the criteria of their evaluation. In contrast, existing games over-visualize the system through explicit status bars, relationship notifications, and comprehensive flowcharts. According to Manokha (2018), the core of the panoptic mechanism relies on the “omnipresence of the inspector ensured by his total invisibility.” In other words, for a system to truly discipline the subject’s mind, the evaluative gaze must remain hidden so that the watched assumes they are being observed at all times. By over-exposing these metrics, however, interactive games allow the “inspector” (the game system) to become visible, thereby breaking the illusion of omnipresent surveillance.

As shown in the flowchart of ‘Detroit: Become Human’ (Figure 5), exposing the entire narrative architecture dissolves the “sustained uncertainty” essential for psychological discipline. When the rules of judgment are laid bare, players cease to internalize the gaze and instead adopt the position of an “external analyst.” As Whitson (2013) notes, such quantified interfaces reduce players to calculable objects who prioritize modifying their behavior based on visible data over internal moral work. Consequently, instead of engaging in spontaneous moral reflection, players focus on strategic optimization, subordinating ethical inquiry to systemic efficiency. Thus, while these systems successfully pressure the player, they ultimately fail to maintain the state of continuous self-regulation required for a true panoptic experience, turning ethical choices into technical problems of system optimization.

4. GAME DESIGN: BUTTERFLY EFFECT

4.1 Game Overview



Figure 3. Title screen of ‘Butterfly Effect’

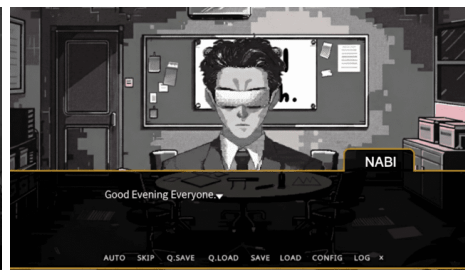


Figure 4. In-game screen of ‘Butterfly Effect’

‘Butterfly Effect’ (Shim, 2024) is a visual novel set within the context of group counseling sessions. Developed using Unity and the UTAGE4 (Unity Text Adventure Game Engine Version 4) dialogue system, the prototype consists of six primary dialogue scenes with 24 choice points. These choices lead to five different endings based on the accumulated ‘Suspicion Parameter’, which further determines the availability of additional final choices. Players experience the game from the perspective of the protagonist, Jiyoung, listening to the stories of various individuals and selecting questions and responses. While the game adopts a dialogue-centered narrative structure on the surface, player choices go beyond mere narrative branching to reflect the player’s subjective judgment within the dialogue. Specifically, the system tracks the player’s inclination to either doubt or accept a character’s statements, and these judgments are quantified through a hidden ‘Suspicion Parameter.’ This hidden metric leads to subtle changes in how characters respond to the protagonist. Ultimately, the accumulation of this parameter dictates the trajectory of the ending, a process that will be analyzed in detail in the following section regarding its design principles.

4.2 Core System: The Suspicion Parameter

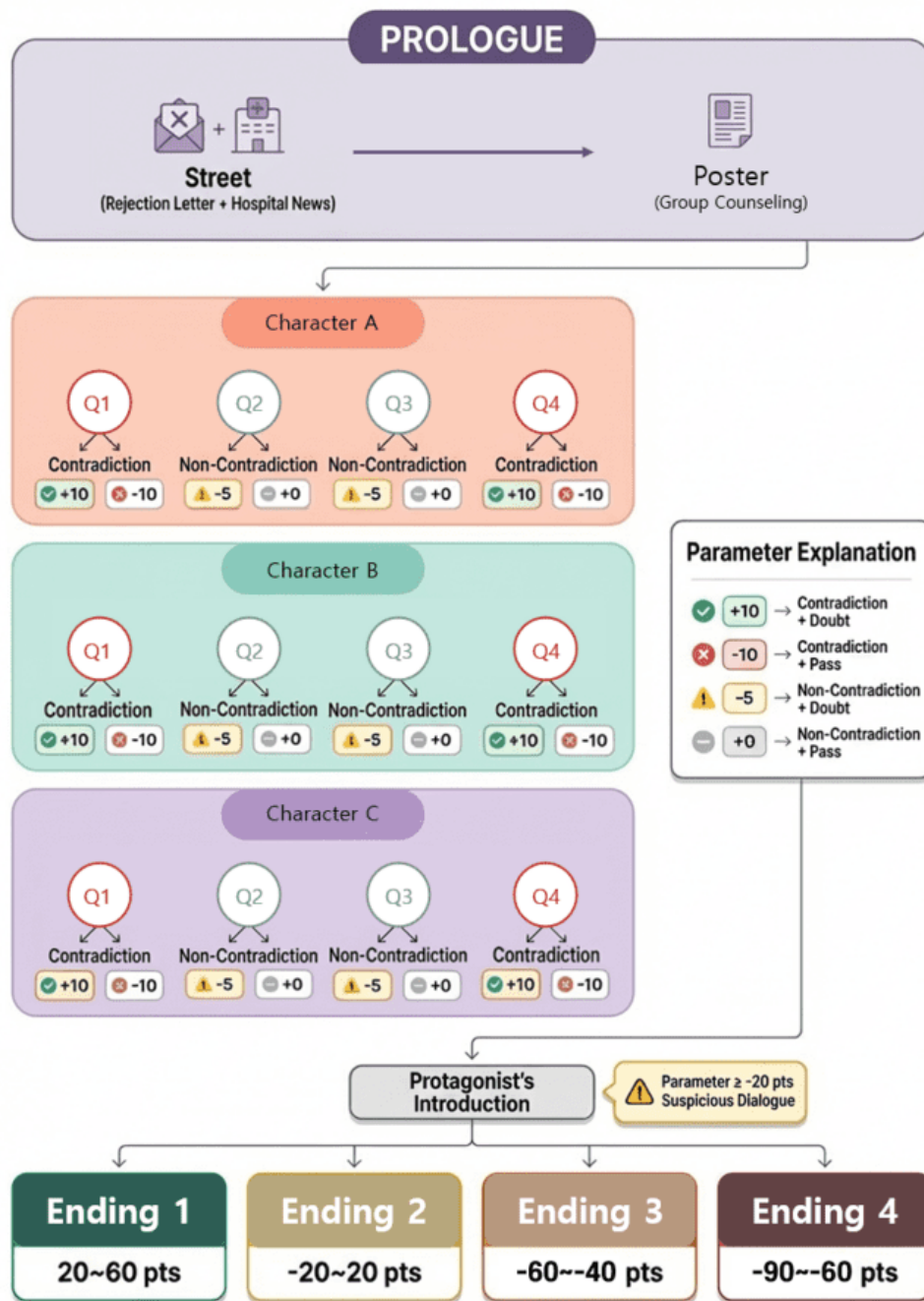


Figure 5. The suspicion parameter system in 'Butterfly Effect'

The core system of *Butterfly Effect* is the 'Suspicion Parameter,' which accumulates player responses internally. While player choices affect this parameter, its value is never directly displayed through the UI, nor are the criteria for its increase or decrease explicitly disclosed. As a result, players cannot clearly determine how their choices are being recorded, nor can they assess the correctness of their decisions through numerical feedback. The 'Suspicion Parameter' thus functions not as a score or reward mechanism, but as an invisible evaluative structure beyond the player's direct perception.

Although this Parameter influences the game's ending, its operation is neither linear nor simple. Both excessively high and excessively low values lead to negative outcomes, and no explicit threshold or correct

range is presented to the player. By obscuring the ‘Suspicion Parameter’, ‘Butterfly Effect’ reproduces a power that is “visible but unverifiable” (Foucault, 2008). This design maintains a state of sustained uncertainty, ensuring that the player remains under the constant pressure of an invisible evaluator without ever being able to confirm the precise criteria of their judgment. This ending structure positions the ‘Suspicion Parameter’ not as a target for optimization, but as an unstable criterion that continuously forces players to reconsider the balance of their decision-making attitudes.

4.3 Changes in Character Responses and Narrative Expression

Players repeatedly choose to doubt or accept character statements, which gradually shifts how characters treat the protagonist. Instead of explicit conflicts, these changes manifest through subtle narrative cues—such as dialogue tone and questioning patterns. Once the ‘Suspicion Parameter’ reaches a threshold, characters begin to insert intrusive probes into daily routines under the guise of concern.

The characters maintain a friendly demeanor, yet an implicit gaze of assessment persists throughout the dialogue. Crucially, the system refrains from explaining the causal links behind these shifts, relying instead on nuanced interactions to instill a persistent sense of ongoing evaluation. This creates an environment where players feel the weight of an invisible gaze even without direct feedback. Consequently, players remain in a state of self-reflexive tension, perceiving character responses not as discrete events but as a pervasive shift in the system’s evaluative attitude.

4.4 The Game Structure as a Panoptic Mechanism

This study proposes a structural model to explain how game systems regulate player judgment, categorized into four distinct stages of panoptic discipline. Stage 1 (Architectural Setup): The game establishes a hidden disciplinary structure by embedding internal evaluation criteria and invisible parameters, creating a state of information asymmetry. Stage 2 (Mechanism of Gaze): The system exercises “unverifiable monitoring.” While it provides visible cues that the player is being watched, it deliberately obscures the actual evaluation results. Stage 3 (Psychological Internalization): This structural opacity leads to a “trap of constant uncertainty.” Deprived of explicit feedback, players begin to internalize the system’s evaluative perspective to alleviate anxiety. Stage 4 (Behavioral Outcome): Finally, players reach a state of self-censorship and regulation. Disciplinary power achieves its automatic functioning as players filter their own thoughts and align their judgment with perceived system norms even before a choice is made.

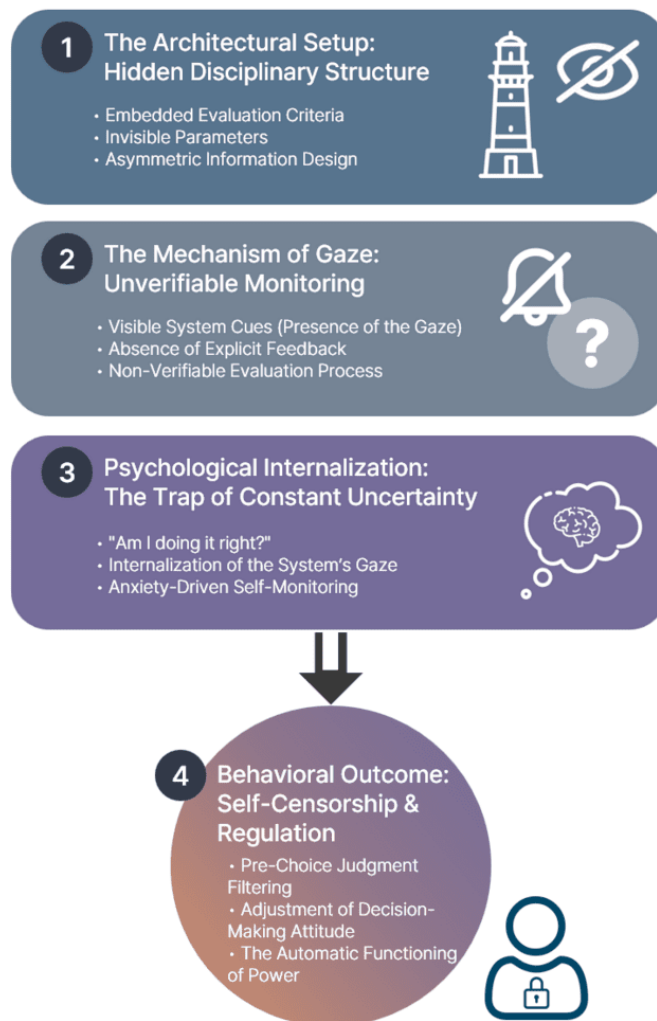


Figure 6. Panoptic mechanism in game design: a four-stage model of internalized discipline

The proposed model is effectively implemented in the design of ‘Butterfly Effect’. In this prototype, the “Suspicion Parameter” operates as an invisible monitoring gaze. By withholding immediate score-based feedback, ‘Butterfly Effect’ shifts the player’s focus from strategic calculation to interpreting subtle narrative cues, such as shifts in character tone. Because the game provides no formal explanation for these shifts, players are forced to navigate a state of sustained uncertainty. Ultimately, this ensures that players do not simply learn game rules, but instead constantly scrutinize and adjust their own decision-making attitudes in response to the invisible, evaluative gaze.

This structural opacity shifts the player’s focus from calculating rewards to interpreting subtle narrative cues. Consequences do not manifest as discrete events or explicit conflicts; rather, they appear as a pervasive shift in the characters’ tone and questioning patterns. For instance, when suspicion reaches a certain threshold, characters maintain a friendly demeanor but begin to insert intrusive probes into the player’s daily routine under the guise of concern. Because the game provides no explanation for these shifts, players are forced to navigate a state of sustained uncertainty. Ultimately, this design ensures that players do not simply learn game rules, but instead constantly scrutinize and adjust their own decision-making attitudes in response to an invisible, evaluative gaze.

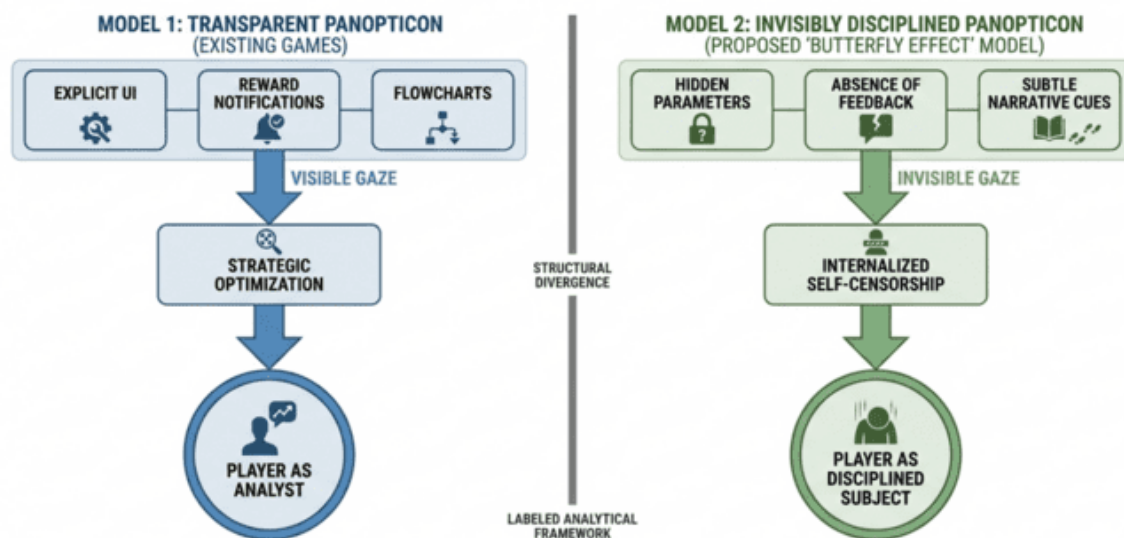


Figure 1. Structural comparison between existing interactive games and ‘Butterfly Effect’

5. CONCLUSION

5.1 Summary and Research Significance

This study investigated the structural power dynamics of choice-based games, focusing on how game systems function as a disciplinary gaze that shapes player judgment. The research began by identifying a convergence of choice in interactive narratives, where players tend to prioritize finding a predetermined correct path. By applying a panoptic perspective, this study reinterpreted this phenomenon as a process of subjectification, where players internalize the system’s gaze and begin to self-censor their decisions.

A key finding of this research is that existing interactive games often render their evaluative criteria highly visible through explicit feedback—such as status notifications and flowcharts. To address this, this study proposed a design framework that sustains the invisibility of the gaze by withholding immediate feedback and maintaining structural uncertainty. Specifically, the research formulated a four-stage model of panoptic discipline—ranging from architectural setup to behavioral outcome—to provide a structured overview of how player uncertainty transitions into internalized self-regulation.

The significance of this study lies in offering a new perspective to interpret player decision-making through the lens of panopticism. Rather than focusing solely on quantifiable data, this research suggests that the sustained uncertainty of evaluation can effectively induce the internalization of power relations, compelling players to regulate their own attitudes in alignment with the invisible system. Ultimately, this work contributes to the field by proposing a design structure where players become the principle of their own subjection, shifting the focus of interactive choice from branching outcomes to the internalized regulation of decision-making attitudes.

5.2 Limitations and Future Work

Despite these structural proposals, this study has several limitations. First, the analysis and design primarily address visual novels and interactive narrative genres. Whether this panoptic structure can be directly applied to other genres or requires significant adaptation remains beyond the current scope. Second, as this study focuses on the structural conditions of power and pre-decision cognition, it does not comprehensively address affective responses such as enjoyment or immersion.

Most importantly, while the prototype was successfully implemented and a four-stage model of panoptic discipline was formulated through this research, this study did not empirically validate the psychological effects on players through surveys or interviews. Accordingly, the objective was not to verify specific behavioral changes, but to propose a structured design framework—specifically the internalized surveillance through information asymmetry—under which such effects may emerge. Future research should employ the developed prototype to investigate how players internally construct evaluative criteria in the absence of explicit feedback. Specifically, future studies will employ post-play surveys to examine how players perceive and internalize the evaluative gaze under different feedback conditions. By comparing self-reported levels of decisional uncertainty, self-censorship, and perceived autonomy between players who experience conventional transparent-feedback structures and those who engage with the panoptic design of ‘Butterfly Effect,’ the research aims to provide initial empirical grounding for the proposed framework. Through such empirical investigation, comparative studies between conventional “transparent” structures and the proposed “panoptic” structures will further clarify how different design conditions shape player judgment, self-regulation, and the internalization of power.

REFERENCES

- Calleja, G., 2009. Experiential Narrative in Game Environments. *Proceedings of the Digital Games Research Association Conference 2009*. London, UK.
- Camingue, J. et al., 2021. What is a visual novel? *Proceedings of the ACM on Human-Computer Interaction*, 5(CHI PLAY), Article 285, pp.1–18. <https://doi.org/10.1145/3474712>
- Carstendottir, E., Kleinman, E. and Seif El-Nasr, M., 2019. Player interaction in narrative games: structure and narrative progression mechanics. *Proceedings of the 14th International Conference on the Foundations of Digital Games (FDG '19)*. New York, USA, pp.1–9, pp. 1–9. <https://doi.org/10.1145/3337722.3337730>
- Cho, G.-H., 2021. Michel Foucault’s theory of power and the discourse of discipline. *Correctional Discourse*, 15(3), 277–301.
- Czauderna, A. and Budke, A., 2020. How Digital Strategy and Management Games Can Facilitate the Practice of Dynamic Decision-Making. *Education Sciences*, 10(4), Article 99.
- Fang, Z. et al., 2021. Interactive movie design based on the game engine technology. *Proceedings of the 2020 4th International Conference on Electronic Information Technology and Computer Engineering (EITCE '20)*. Xiamen, China, pp.1029–1033.
- Fawcett, C., 2020. Consumed by Guilt: Retribution and Justice in Until Dawn. *Interdisciplinary Justice Research*, 9, pp.86–109.
- Ferchaud, A. and Oliver, M. B., 2019. It’s my choice: The effects of moral decision-making on narrative game engagement. *Journal of Gaming & Virtual Worlds*, 11(2), pp. 101–118.
- Foucault, M., 2008. ‘Panopticism’ from Discipline & Punish: The Birth of the Prison. *Race/Ethnicity: Multidisciplinary Global Contexts*, 2(1), pp. 1–12.
- Foucault, M., 2019. *Discipline and Punish: The Birth of the Prison*. Penguin Books Limited, London, UK.
- Kim, E. and Lee, S., 2016. Gaming as Media Use: from “What They Use” to “How They Use”. *Korean Journal of Journalism & Communication Studies*, 60(5), pp. 261–291.
- Kirginas, S. and Gouscos, D., 2018. A Model for the Quantitative Assessment of Freedom of Choice in Adventure Digital Games. *Journal of Digital Media & Interaction*, 1(2), pp. 7–24.
- Liagkou, V. et al., 2021. On the undecidability of the Panopticon detection problem. arXiv preprint arXiv:2107.05220.
- Manokha, I., 2018. Surveillance, panopticism, and self-discipline in the digital age. *Surveillance & Society*, 16(2), pp.219–237.
- Moallem, J.D. and Raffé, W.L., 2020. A review of agency architectures in interactive drama systems. *Proceedings of the 2020 IEEE Conference on Games (CoG)*, Osaka, Japan, pp.1–8. <https://doi.org/10.1109/CoG47356.2020.9231655>
- Oliver, G., 2020. Renegade or Paragon?: Categorizing Narrative Choice in Video Game Storylines. *Dialogue: The Interdisciplinary Journal of Popular Culture and Pedagogy*, 7(1), Article 9.
- Park, S., 2022. A study on game narrative and players’ experience. *Multicultural Contents Studies*, 40, pp. 143–168.
- Quantic Dream, 2018. *Detroit: Become Human* [video game]. Sony Interactive Entertainment, PlayStation 4.
- Rogers, R., 2017. The motivational pull of video game feedback, rules, and social interaction: Another self-determination theory approach. *Computers in Human Behavior*, 73, pp.446–450. <https://doi.org/10.1016/j.chb.2017.03.048>.

- Ryan, M. et al., 2023. The effect of morality meters on ethical decision making in video games: A quantitative study. *Computers in Human Behavior*, 142, 107623. <https://doi.org/10.1016/j.chb.2022.107623>
- Salen, K. and Zimmerman, E., 2004. *Rules of play: Game design fundamentals*. MIT Press, Cambridge, MA, USA.
- Shim, A. (2024). *Butterfly Effect* [Video game]. Available at: <https://skku-gamecenter.itch.io/butterfly-effect> (Accessed: 3 April 2026).
- Supermassive Games, 2015. *Until Dawn* [video game]. Sony Computer Entertainment, PlayStation 4.
- Telltale Games, 2016. *The Walking Dead: A New Frontier* [video game]. Telltale Games.
- Wang, H., and Sun, C.-T., 2011. Game reward systems: Gaming experiences and social meanings. *Proceedings of the Digital Games Research Association Conference 2011*. Hilversum, The Netherlands.
- Whitson, J.R., 2012. *Game Design by Numbers: Instrumental Play and the Quantitative Shift in the Digital Game Industry*. PhD thesis. Carleton University, Ottawa, Canada.
- Whitson, J.R., 2013. Gaming the Quantified Self. *Surveillance & Society*, 11(1/2), pp. 163-176.
- Wolf, M.J.P. and Perron, B. (eds.), 2003. *The Video Game Theory Reader*. Routledge, New York, USA.
- Yin, M. and Xiao, R., 2022. How Should I Respond to “Good Morning?”: Understanding Choice in Narrative-Rich Games. *Proceedings of the 2022 ACM Designing Interactive Systems Conference (DIS '22)*. Virtual Event, Australia, pp. 1–17. <https://doi.org/10.1145/3532106.3533459>